Hong Liu 刘弘



Ph.D. student, Epidemiology and Health Statistics, School of Public Health, Xiamen University (since Sept 2026)

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I trained in preventive medicine, took a second degree in mathematical statistics, and then did a master's in computer science and technology. My research is infectious-disease forecasting: using surveillance data and mathematical models to estimate how an outbreak will develop and how much control measures help. I am first author of two journal papers and one preprint under review, and co-author of six more papers.

EDUCATION

Xiamen University, School of Public Health — Ph.D. student, Epidemiology and Health Statistics Sept 2026 – present

Supervisor: Prof. Tianmu Chen. Topics: infectious-disease forecasting and early warning; seasons when several influenza strains circulate at once (co-circulation); outbreak transmission models.

M.Sc., Computer Science and Technology — joint training programme of Macau University of Science and Technology (Faculty of Innovation Engineering) and Xiamen University Sept 2024 – 2026

GPA 3.63/4; Full Scholarship and living allowance. Supervisor: Prof. Chitin Hon; Lab director: Academician Nanshan Zhong; Xiamen University supervisor: Prof. Tianmu Chen.

Xiamen University — B.Med., Preventive Medicine (School of Public Health); B.Sc., Mathematical Statistics (second degree, Wang Yanan Institute for Studies in Economics) Sept 2019 – June 2024

RESEARCH

How fast chikungunya spread in Foshan in 2025, and how well control worked Co-first author (listed first) · published 2025 – 2026

- **Problem:** Chikungunya is spread by mosquitoes. During the 2025 Foshan outbreak the city tightened control in three phases. The questions were how fast the virus spread and how much each phase helped.
- **What I did:** Fitted the same case data with two different kinds of model — a standard differential-equation model and a Petri net, which describes infection and recovery as discrete events — and used each to check the other. I built the three control phases into the models and varied the parameters one by one to find which mattered most.
- **Result:** Both models fit daily new cases with $R^2 \approx 0.95$; the predicted peak came 3 days early and about 6% low. The early transmission figure (how many people one patient infects with no control in place) was 14.67 and 13.90, so the models agree. After phase three the effective number fell to 0.059, far below 1, meaning transmission was stopped. The acute-phase recovery rate (γ) mattered most: it explains 96.7% of the variation in the early transmission figure (Sobol index 0.9672). Published in *BMC Public Health* (2026).

MAESTRO: forecasting flu from several data sources First author · preprint, under journal review 2025

- **What I did:** Case-only forecasts lag and miss unusual seasons, so I combined three data sources: clinic reports of flu-like illness, Google search activity and weather. The model splits the case curve into long-term trend, seasonal pattern and short-term swings and learns each one separately. Each forecast comes with a likely range.
- **Result:** On more than 11 years of Hong Kong flu data (COVID-19 period excluded), $R^2 = 0.956$, meaning the forecasts explain about 96% of the real ups and downs. Code and full reproduction steps are open source (arXiv:2509.08578).

How flu seasons change when several virus strains circulate together Co-first author (listed first) · published 2025 – 2026

- **Problem:** When influenza A and B strains circulate together, the peak can come early, come late or change shape, so forecasts based on past seasons go wrong.
- **What I did:** Built 323 weekly time series from flu surveillance in northern and southern China, 2011–2025. Split each curve into trend, seasonal pattern and short-term swings, extracted timing and cycle features, and compared “several strains at once” with “one strain dominates”. Then measured how closely flu moved with weather and other factors (63,308 variable pairs). I designed the method, wrote all the analysis code and ran the checks.
- **Result:** When several strains circulate, the seasonal pattern weakens (seasonal strength down 28%). In the north, flu still follows the climate; in the south it “decouples” — e.g., H3N2’s alignment with soil moisture dropped from 0.45 to 0.07. Published in *npj Systems Biology and Applications* (2026); code open source (MultiCoPat).

Survey study of burnout among health-care workers Second author · under review Nov 2024 – Nov 2025

- **Problem:** Burnout is common among health-care workers. The study asked whether psychological resilience protects against it, and what roles work stress and trust in the employer play.
- **What I did:** Cleaned the survey data, checked the questionnaires’ reliability and validity, tested how the factors connect with structural equation modelling, and checked indirect effects with 5,000 bootstrap resamples.
- **Result:** Higher resilience went with less burnout ($\beta = -0.283$, $p < 0.001$). Part of this works through lower work stress (indirect effect −0.217), and trust in the organisation changes how strong the link is ($p < 0.01$). Submitted to *Health Soft Science* (卫生软科学).

Undergraduate project: can China eliminate HIV/AIDS? Modelling and coding May – Aug 2023

- Wrote a scraper to collect public HIV data; forecast future infections and deaths with time-series and neural-network models; built a transmission model based on disease stages and an individual-level simulation to test interventions in small outbreaks.

Forecasting dengue mosquito density (Breteau Index) in Fujian Second author · preprint Sept 2023 – 2024

- The Breteau Index (mosquito larvae around homes) is a standard dengue-risk indicator. Combined decision trees, random forests, support vector machines and a neural network using weather and habitat data; accuracy on data the model had not seen was 0.865. Preprint on Research Square (2024).

ENGINEERING PROJECTS

Infectious-disease data collection, database and public dashboard Built alone Oct 2024 – present

- **Problem:** Outbreak data sit on many websites in different formats, so the group downloaded and tidied them by hand.
- **What I did:** Wrote scrapers that download data from the Public Health Science Data Center by year, disease, and region or age group, and merge monthly files into yearly totals; also collected local CDC web pages. Cleaned data go into a database and onto a web map, updated

daily; failed downloads retry and resume. In 2026 I added EPIC, a public dashboard that gathers outbreak events from health-agency notices, ProMED and news feeds, each record showing its source and update time.

- **Result:** The group can look up data directly by region, disease and date. Two parts are public on GitHub: the region/age scraper phsciencedata_crawler_region (5 stars) and the EPIC dashboard epic-site-pages.

Leak-risk simulation for a high-containment biosafety lab (with Guangzhou Medical University Tansha Lab) June 2024 – Apr 2025

- **Problem:** A biosafety level 3 (P3) lab handles dangerous pathogens; the team needed to know in advance where one could escape and by what route.
- **What I did:** Built a 3D model of the lab in Unity that simulates the routes of people and materials and the indoor airflow, and produces risk heat maps. I broke possible accidents down into step-by-step chains of causes and scored them by likelihood × severity. The assessment workflow and settings interface were built with Python, Flask and MySQL.
- **Result:** Delivered a demo project, operating procedures, assessment-form templates and a draft risk-assessment guide. Co-wrote annual research reports and a draft patent on simulating urban virus transmission routes.

AI protein-design workflow Aug 2024 – Mar 2025

- Set up RFDiffusion, Rosetta, AlphaFold 3 and related tools to design proteins that bind a target protein and predict whether they are stable; turned the steps into reusable scripts.

OPEN-SOURCE CONTRIBUTIONS (PERSONAL GITHUB WORK, NOT EMPLOYMENT)

- **bibverify** (my own tool): checks that the references in a paper are real and correct. It verifies DOI, year and authors, fills in missing details and removes duplicates. Published on PyPI; about 99 stars on GitHub.
- **MiniMax-AI / MiniMax-Code-Plugins** (official plugin library for MiniMax's AI coding assistant): 6 plugins merged, Aug–Sept 2026, including the openscience research workflow; mcodeforlegal, for legal work in mainland China (contract review, litigation, compliance) with checks against statute text; and plugins for protein-binder design, document handling and command-line tool links.
- **hetaoBackend / MiniMax-Code-Plugins**: 3 plugins merged in Aug 2026 (search-first, verification-loop, mcp-server-patterns).
- **lidge-jun / opencodex**: #6578 merged in Oct 2026. It fixed tests that interfered with each other and hung when child processes did not exit.

PUBLICATIONS

1. **First author** (co-first). Dual-Model framework for CHIKV transmission modeling: ODE and Petri Net Analysis of the 2025 Foshan outbreak. *BMC Public Health*, 2026, 26. DOI: 10.1186/s12889-026-28110-9
2. **First author** (co-first). Spatiotemporal instability of influenza seasonality during viral co-circulation. *npj Systems Biology and Applications*, 2026, 12. DOI: 10.1038/s41540-026-00729-9
3. **First author**. Multi-modal Adaptive Estimation for Temporal Respiratory Disease Outbreak (MAESTRO). arXiv:2509.08578, 2025 (under journal review).
4. Co-author (7/11). Dengue transmission dynamics in China's border regions adjacent to Myanmar. *Infectious Diseases of Poverty*, 2026. DOI: 10.1186/s40249-026-01440-x
5. Co-author (7/14). Temporal shifts in 24 notifiable infectious diseases in China before and during the COVID-19 pandemic. *Nature Communications*, 2024. DOI: 10.1038/s41467-024-48201-8
6. Co-author. Transmissibility of the variant of concern for SARS-CoV-2 in six regions. *Heliyon*, 2024. DOI: 10.1016/j.heliyon.2024.e32164
7. Co-author (2/5). Impact of long-term care insurance on the physical and mental health of older adults in China. *Medicine and Society* (医学与社会), 2024. DOI: 10.13723/j.yxysh.2024.08.011
8. Co-author (4/13). Assessment of transmissibility and measures effectiveness of SARS in 8 regions, China, 2002–2003. *Frontiers in Cellular and Infection Microbiology*, 2023.
9. Co-author (7/11). Advancements in Defining and Estimating the Reproduction Number in Infectious Disease Epidemiology. *China CDC Weekly*, 2023.
10. Co-author (2/10). Three machine-learning models to predict dengue vectors Breteau Index based on meteorology and biotope in Fujian China. Research Square preprint, 2024.
11. Co-author (2/6). Psychological resilience, occupational stress, organizational trust, and burnout among healthcare workers. *Health Soft Science* (under review).

COMPETITIONS AND AWARDS

- 2025.06 Second Prize, 20th Citi Cup Financial Innovation Application Competition (daily yuan exchange-rate forecast and risk-alert dashboard)
- 2025.06 Regional First Prize and National Third Prize, 15th MathorCup Mathematical Application Challenge, graduate group (listed 1/3)
- 2024.12 Third Prize, 21st China Graduate Mathematical Contest in Modeling (Huawei Cup) (listed 1/3)
- 2024.12 Project Innovation Award, 13th "Win in Guangzhou" Entrepreneurship Competition, Hong Kong–Macao Division
- 2024.04 National Second Prize, East China Regional First Prize and Shanshu Operations Research Fund Award, 14th MathorCup (project lead)
- 2023.08 National Bronze Award; Preventive Medicine Track First Prize, 9th National Undergraduate Basic Medical Innovation Research and Experimental Design Forum
- 2023.04 Honorable Mention, Mathematical Contest in Modeling (MCM)
- 2022.12 Third Prize, 10th Xiamen University Public Economics and Policy Paper Competition
- 2021 Gold Award, 5th China Youth Volunteer Service Project Competition

SKILLS AND OTHER

- **Tools and methods:** Python, R, Stata, C++, JavaScript; PyTorch, scikit-learn; Linux, Git, LaTeX, Flask, MySQL, Unity; transmission models, time-series forecasting, structural equation modelling.
- **Languages:** Mandarin (native); English (CET-4, CET-6, IELTS Academic 6.5), reads and writes research papers. **Student roles (2019–21):** deputy head, Innovation Service Dept., XMU Student Sci-Tech Entrepreneurship and Employment Service Center.